

Thaisausage ERP Integration Operations Guide

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Audience: Thaisausage IT / operations / deployment team
Domain: `https://thaisausage.krs.co.th`

1. System purpose

ระบบนี้เป็น middleware ระหว่าง ERP ที่เก็บข้อมูลใน SQL Server กับ eVRP การออกแบบใช้ Schedule ดึงเฉพาะ SO ที่ Approve แล้วด้วย SELECT-only ก่อน map เป็น JSON และส่งไป eVRP เมื่อเปิด live mode หลังผ่านการอนุมัติและทดสอบ

```
Schedule -> SQL Server SELECT (IsApprSo = 1) -> mapper -> submission guard -> eVRP
eVRP D0 -> /vrp/do-received -> do_receipts staging -> operator review
```

2. Production deployment currently installed

Component	State
VPS	Linux host <code>krs-cloud</code>
Runtime	Docker 29.8 / Compose 5.5
App	Python container, non-root <code>appuser</code> , health check
Reverse proxy	Caddy 2 with automatic HTTPS
Public URL	<code>https://thaisausage.krs.co.th</code>
Persistent state	Docker volume for SQLite
Current mode	<code>dry_run=true</code> , <code>sync.enabled=false</code>

GET `/health` is public and reports process state only; it does not prove SQL/eVRP readiness. `/docs` and `/openapi.json` expose contracts but no credentials or payload history.

3. Configuration ownership

Application secrets live only in `/opt/thaisausage/.env` on the VPS with mode `600` . The file is not in GitHub. Keep it separate from `deploy/vps.env` , which contains VPS connection details.

Required application variables:

Variable	Purpose
THAISAUSAGE_API_KEY	authenticates ERP to middleware
VRP_TOKEN	authenticates middleware to eVRP as X-Token
ERP_SQLSERVER_CONNECTION_STRING	server/database/encryption portion of ODBC config
ERP_SQL_USER / ERP_SQL_PASSWORD	SQL Server read-only credential
ERP_SQLSERVER_WRITE_CONNECTION_STRING	ODBC config for the DO write account (unused while <code>do_write.enabled=false</code>)
ERP_SQL_WRITE_USER / ERP_SQL_WRITE_PASSWORD	least-privilege DO write credential; must differ from the read-only pair

`sqlserver.approved_orders_query` is the only query used by scheduled SO sync. It must be a reviewed, parameter-free SELECT that filters only ERP-approved SOs. The application refuses to run a query that does not contain the `IsApprSo` predicate, and refuses any statement that is not a plain SELECT (INSERT, UPDATE, DELETE, MERGE, EXEC, ALTER, DROP, TRUNCATE and INTO are rejected).

`config/local.json` never holds secrets; it only names environment variables. It is listed in `.gitignore` and `deploy/.dockerignore`, the image ships only `config/example.json`, and the real file is mounted read-only at runtime. `do_write.enabled` must stay `false` in every deployed release.

SQL password is appended by the connector at runtime and is not stored in the base connection string. Use `Encrypt=yes;TrustServerCertificate=no` with a trusted certificate. Do not log the connection string.

`config/local.json` is a non-secret runtime file. It must define `source`, SQL settings, approved query set, mapping, `dry_run` and `sync.enabled`. Keep `dry_run=true` and `sync.enabled=false` until read-only SO verification is complete.

4. API responsibilities

Endpoint	Responsibility	State effect
POST /api/v1/erp/hooks/order-ready	legacy compatibility receiver	stores <code>erp_hooks</code> ; not used by scheduled SO sync
POST /api/v1/erp/orders	direct standard SO ingestion	validates and submits through guard
POST /api/v1/erp/pull	legacy/configured REST pull	reads configured ERP REST endpoint
POST /api/v1/erp/do-received	DO staging	adds <code>do_receipts</code> , never writes ERP
POST /api/v1/vrp/do-received	eVRP DO callback	adds <code>do_receipts</code> under <code>source=eVRP</code> , never writes ERP
GET /api/v1/submissions/{request_id}	delivery status	read-only
GET /health	liveness	read-only

The scheduled worker runs every `sync.interval_seconds` when `sync.enabled=true` . It executes only the reviewed `sqlserver.approved_orders_query` , which must filter SOs with `IsApprSo = 1` , maps the returned rows and uses the same submission guard. The old webhook is no longer the primary trigger. Each cycle is bounded by the approved query and already-claimed SOs are skipped.

5. SQL Server safety contract

The source credential is SELECT-only. The connector:

- lazily loads `pyodbc` and Microsoft ODBC Driver 18;
- requires a configured, reviewed query;
- permits only statements beginning with `SELECT` ;
- rejects write/side-effect keywords including INSERT, UPDATE, DELETE, MERGE and EXEC;
- binds order parameters instead of concatenating values;
- never accepts SQL text from an HTTP request;
- does not use `NOLOCK` or alter SQL Server settings.

During all tests, only SOs explicitly approved by the user may be selected from real ERP. All other data, delivery, retry, callback and error tests use mocks/staging. No ERP DDL or mutation is allowed.

6. Local state model

The current foundation creates:

```
submissions(request_id, payload_hash, state, result, created_at)
order_claims(order_no, request_id)
erp_hooks(event_id, event_type, source_id, company_id, order_no, changed_at, state,
received_at)
do_receipts(receipt_key PK, source, receipt_id, do_no, payload_hash, payload, state,
received_at)
do_writes(receipt_key PK, source, receipt_id, do_no, transaction_no, payload_hash, state,
reason, updated_at)
```

The current implementation protects request/event/receipt identity and stores remote VRP references when present. It does not yet provide a durable outbox with immutable payload bytes, attempt history, checkpoint leases or operator replay. Those are required before unattended production sync.

7. Operating procedures

Check service

```
cd /opt/thaisausage
docker compose -f deploy/docker-compose.yml ps
docker compose -f deploy/docker-compose.yml logs --tail=100 thaisausage
curl -fsS https://thaisausage.krs.co.th/health
```

Inspect Swagger

Open <https://thaisausage.krs.co.th/docs> and use the OpenAPI contract. Authenticated operations require the internal API key. Never paste a real key into a shared screenshot.

Pre-deploy verification

Run on the build machine before touching the VPS:

```
python3 -m unittest discover -s tests -v
python3 -m json.tool config/example.json >/dev/null
python3 -m json.tool docs/openapi.json >/dev/null
git diff --check
git status --short          # .env, deploy/vps.env and config/local.json must never appear
```

Then confirm on the VPS copy of `config/local.json`:

- `[] dry_run` is the approved value for this release
- `[] sync.enabled=false` unless the SQL sync gate has been passed
- `[] do_write.enabled=false` — no release enables the ERP DO writer

- [] `sqlserver.approved_orders_query` is the reviewed query filtering `IsApprSo = 1`
- [] `/opt/thaisausage/.env` still has mode `600` and is outside Git
- [] SQLite volume backed up (see backup command in the system manual)

After `docker compose up -d`, check `/health`, `/docs`, `/openapi.json`, container logs, and post one mock DO to `/api/v1/vrp/do-received` expecting `202` with `erp_write: false`.

Rollback

Rolling back code is safe; rolling back the database is not, because remote systems may already hold accepted requests.

```
cd /opt/thaisausage
git log --oneline -5          # pick the last known-good commit
git reset --hard <known-good-commit>
docker compose -f deploy/docker-compose.yml build
docker compose -f deploy/docker-compose.yml up -d
curl -fsS https://thaisausage.krs.co.th/health
```

Rules:

- Keep the SQLite volume as it is. Never restore an older database to "undo" sends: `order_claims`

would be lost and the next cycle could submit the same SO to eVRP twice.

- To stop outbound activity immediately, set `sync.enabled=false` and restart; staging endpoints stay up.
- If a release is rolled back while submissions are `sending` or `needs_review`, reconcile those in eVRP

by business reference before enabling sync again.

- Record the rolled-back commit, the reason and the state counts in the phase report.

Update an image

```
cd /opt/thaisausage
git fetch origin main
git reset --hard origin/main
docker compose -f deploy/docker-compose.yml build
docker compose -f deploy/docker-compose.yml up -d
docker compose -f deploy/docker-compose.yml ps
```

Before update: back up the SQLite volume, verify `.env` remains outside Git, and keep sync disabled. After update: check health, logs, OpenAPI and a mock webhook/DO request.

Pause live behavior

Set `sync.enabled=false` and restart the app. If live sending is enabled, pause the process before restoring an old database backup. Reconcile accepted remote requests first; restoring old claims can cause duplicate attempts.

8. Monitoring and alerts

Minimum checks:

- HTTPS certificate and Domain availability
- app/Caddy container health
- SQL connection/query latency once enabled
- oldest queued/review hook age
- `needs_review`, `review`, `rejected` counts, plus scheduled-cycle states `already_sent`, `prior_attempt_needs_review` and `detail_item_code_missing`
- SQLite volume free space and backup freshness
- authentication failures and eVRP upstream errors

Do not log API keys, SQL passwords, full SO payloads, customer phone numbers or addresses. Use request/event IDs and redacted error codes for correlation.

9. DO staging and accounting boundary

`/api/v1/erp/do-received` is an observation/staging receiver. It never inserts, updates or deletes ERP data. The stored payload is for format discovery and reconciliation only. A future accounting export or ERP write requires a separate vendor contract, permission, idempotency key, reversal rule and approval.

COD callback from eVRP also requires an agreed ingress authentication method and duplicate identity. Do not enable a production callback route merely because the JSON shape is known.

DO writer (`do_write`) — **CODE DONE, disabled**

Since 16-09-26 the codebase contains a writer for `tbl_D0hdr / tbl_Dodtl` (`thaisausage/do_writer.py`). It is off by default and has never written to ERP. Field mapping is still incomplete; see [DO insert contract](do-insert-contract.md).

Operator rules:

- Keep `do_write.enabled=false` on the VPS. Deploy the code with the flag off; enabling is a separate approved change.
- The writer refuses to start if its credential environment variables reuse the read-only pair.

- The eVRP callback calls the writer only when `do_write.enabled=true` and the payload carries a

`transaction_no`. Staging always answers 202; the writer outcome is reported in `do_write` and `erp_write` is true only for `inserted`. The ERP-side DO path never calls the writer.

- Writer states are recorded in `do_writes`: `preview`, `disabled`, `rejected`, `inserted`, `needs_review`.
- `inserted` and `needs_review` are never rewritten automatically. Investigate `needs_review` in ERP by

`TransactionNo` / `DoNo` before any manual action; a commit timeout may mean the rows already exist.

- Duplicate protection: `receipt_id` primary key plus unique `do_no` and `transaction_no`.

10. Release gates

Gate A — read-only SQL discovery

- ☐ ODBC driver and pyodbc verified on runtime image
- ☐ SQL version, schema, source keys and timezone recorded
- ☐ Read-only service identity verified
- ☐ Approved SO sample scope recorded
- ☐ Query plan and row counts recorded without mutation

Gate B — mapping and dry-run

- ☐ Header/detail/customer joins match ERP
- ☐ COD, credit, gift, Thai text, decimal and multiple-line cases match
- ☐ `dry_run=true` preview has no outbound eVRP request
- ☐ No source checkpoint advances during preview

Gate C — controlled eVRP UAT

- ☐ eVRP test appointment and test token confirmed
- ☐ Mock tests and local recovery tests pass
- ☐ Controlled test records approved by business owner
- ☐ Remote identifiers and amounts reconcile with source
- ☐ User confirmation recorded

Gate D-DO — ERP DO write (separate from eVRP release)

- ☐ DBA schema, constraints and mapping matrix signed
- ☐ Dedicated write credential issued and scoped to the two approved tables
- ☐ Approved test database or explicitly approved DO record available
- ☐ Mock transaction/rollback/duplicate tests pass
- ☐ Controlled UAT insert reconciled: Header count, Detail count and shared `TransactionNo`
- ☐ No unrelated ERP rows changed
- ☐ ERP owner and business/data owner sign-off recorded before `do_write.enabled=true`

Gate D — live operation

- ☐ Backup/restore drill passed
- ☐ TLS/firewall/secret rotation documented
- ☐ Runbook and escalation owners assigned
- ☐ SQL sync enabled only for agreed scope
- ☐ First operating cycle observed before expansion

11. Recovery rules

Reading a refusal from eVRP

A refused submission keeps the upstream explanation, so no second send is needed to learn why:

```
docker compose -f deploy/docker-compose.yml exec -T thaisausage python -c "
import sqlite3, json
db = sqlite3.connect('data/integration.sqlite3')
for rid, result in db.execute('SELECT request_id,result FROM submissions WHERE
state="needs_review" ORDER BY created_at DESC LIMIT 5'):
    r = json.loads(result)
    print(rid, r.get('reason'))
    print(' ', r.get('upstream_error', '-')[:400])"
```

Known refusals and their owner: `PICKUP_HUB_NOT_FOUND` and `CUSTOMER_MASTER_REQUIRED` are eVRP master-data tasks; `REQUEST_ID_CONFLICT` means the same identity was reused with different data. Clearing an `order_claims` row to allow a fresh attempt is permitted only after a definitive refusal (422/409), never after a timeout whose outcome is unknown.

`needs_review` means the outcome may be unknown. Check eVRP using its business reference before replay. Never delete `order_claims` to force a resend. A changed SO after send becomes an

amendment review item; there is no documented update/cancel contract in the current integration package.

For a failed deployment, pause sender, preserve the database volume and inspect the last request/event IDs. Do not roll back by silently replacing the database or resetting checkpoints.

12. Document references

- [ERP Integration Guide](ERP_Integration_Guide.md)
- [OpenAPI JSON](openapi.json)
- [ERP source contract](erp-source-contract.md)
- [Deployment Compose](../deploy/docker-compose.yml)
- [SQL Server integration plan](../process/features/erp-sqlserver/active/ERP_SQLSERVER_PLAN_12-09-26.md)

This document describes the current deployed foundation. It does not claim that SQL mapping, live eVRP delivery, accounting posting or automatic recovery has passed UAT.